

D276

Web Development Foundations

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A stylesheet written by a design agency and loaded from a CDN has these rules in order: `#intro { color: red; }` then `p { color: blue; }` then `.lead { color: green; }`. The element is `<p id=intro class=lead>`. What color is the text?

- A. Red
- B. Blue
- C. Green
- D. Black, because conflicting rules cancel out

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

Text is colored by the rule `.menu .item { color: gray; }`. Which new selector, with the same property, would override it?

- A. `a`, placed later in the file
- B. `div ul li a`
- C. `#main a`
- D. `.item`

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A product card for a shoe store has `width: 300px; padding: 20px; border: 5px solid; margin: 10px; background-color: #f5f5f5;` and the default box-sizing. How wide is the card from outer border edge to outer border edge?

- A. 300px
- B. 350px
- C. 370px
- D. 325px

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

An element has box-sizing: border-box; width: 300px; padding: 20px; border: 5px solid. How wide is its content area?

- A. 350px
- B. 260px
- C. 250px
- D. 300px

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A 12-page site that sells handmade candles has a bar of links to Home, Shop, About, and Contact at the top of every page. Which element best marks up that group of links?

- A. <nav>
- B. <header>
- C. <div class=nav>
- D. <aside>

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A page with a dark theme has New styled with .badge { width: 200px; background: gold; }. The background shows, but the width is ignored. Why?

- A. Width must be given in percent.
- B. The width needs !important.
- C. A span can never be given a size.
- D. span is inline; set display: inline-block or block.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A site hosted on a \$6-a-month shared plan has the page /products/shoes.html and the image /images/logo.png. Which relative path in the page's img tag loads the logo?

- A. ../../images/logo.png
- B. file:///C:/site/images/logo.png
- C. images/logo.png
- D. ../images/logo.png

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

The page `/blog/2026/post.html` contains an image with `src ../img/a.png`. Which file does the browser request?

- A. `/img/a.png`
- B. None; relative paths can never go up a folder.
- C. `/blog/2026/img/a.png`
- D. `/blog/img/a.png`

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A login form for a clinic portal has a blue submit button and fields for username and password. The site uses HTTPS. Which method should the form use, and why?

- A. GET, because it is faster.
- B. POST, so the password is not placed in the URL, history, or logs.
- C. Either; the two methods always behave the same.
- D. GET is fine because the site uses HTTPS.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

Analytics from last quarter show most visitors to a recipe site use phones. The designer wants a one-column layout only on screens 600px wide or narrower. Which media query fits?

- A. `@media (min-width: 600px)`
- B. `@media print`
- C. `@media (max-width: 600px)`
- D. `body { width: 600px; }`

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

A stylesheet has `@media (min-width: 768px) { .sidebar { display: block; } }` and `.sidebar` is hidden otherwise. At which window width is the sidebar shown?

- A. 1024px
- B. Any width, because media queries are ignored without a framework
- C. 500px
- D. Only at exactly 768px

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A nonprofit’s annual report page shows a 400 KB PNG bar chart revealing that donations rose 20% from 2025 to 2026. The chart is the only place this fact appears. Which alt text is best?

- A. No alt, but a title attribute with the description
- B. “Bar chart: donations rose 20% from 2025 to 2026”
- C. An empty alt attribute
- D. “chart.png”

Target word	The one fact that decides	Answer	☐ sure ☐ unsure ☐ guess

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

A stylesheet ~~written by a design agency and loaded from a CDN~~ has these rules in order: #intro { color: red; } then p { color: blue; } then .lead { color: green; }. The element is <p id=intro class=lead>. What color is the text?

Target word: winning color **Decides:** The ID selector has the highest specificity.

A	Correct	#intro outranks .lead and p regardless of order.
B	Inverted operation	The element selector is the weakest.
C	Confusion set	Last-rule-wins applies only to equal specificity.
D	Absolute language	Conflicts resolve by specificity; they never cancel.

2. Answer C Concept 1 • transfer

Text is colored by the rule .menu .item { color: gray; }. Which new selector, with the same property, would override it?

Target word: selector with higher specificity **Decides:** One ID outranks any number of classes.

A	Adjacent category	Order only breaks ties; (0,0,1) is lower.
B	Confusion set	Four elements (0,0,4) still lose to two classes.
C	Correct	An ID gives (1,0,1), above (0,2,0).
D	Dropped element	One class (0,1,0) is weaker than two.

3. Answer B Concept 2

A product card ~~for a shoe store~~ has width: 300px; padding: 20px; border: 5px solid; margin: 10px; ~~background-color: #f5f5f5;~~ and the default box-sizing. How wide is the card from outer border edge to outer border edge?

Target word: border-box width **Decides:** $300 + 2 \times 20 + 2 \times 5$.

A	Confusion set	That is border-box behavior, not the default.
B	Correct	Content 300 + padding 40 + border 10.
C	Adjacent category	Adds margin, which sits outside the border.
D	Dropped element	Counts padding and border on one side only.

4. Answer C Concept 2 • transfer

An element has box-sizing: border-box; width: 300px; padding: 20px; border: 5px solid. How wide is its content area?

Target word: content width **Decides:** Border-box width includes padding and border.

A	Inverted operation	Adds instead of subtracting.
B	Dropped element	Subtracts padding but not the border.
C	Correct	$300 - 40 - 10$.
D	Wrong stem number	The declared width, which includes padding and border here.

5. Answer A Concept 3

A ~~12-page~~ site ~~that sells handmade candles~~ has a bar of links to Home, Shop, About, and Contact at the top of every page. Which element best marks up that group of links?

Target word: semantic element for main links **Decides:** It is the primary site navigation.

A	Correct	nav marks major navigation blocks.
B	Confusion set	A header may contain nav, but it is not the link group itself.
C	Adjacent category	Looks the same but carries no meaning for assistive tech.
D	Confusion set	For side content loosely related to the page.

6. Answer D Concept 4

A page ~~with a dark theme~~ has `<span class=badge>New</span>` styled with `.badge { width: 200px; background: gold; }`. The background shows, but the width is ignored. Why?

Target word: why width fails **Decides:** span is inline.

A	Adjacent category	Units are not the problem.
B	Confusion set	No other rule is overriding it.
C	Absolute language	Changing display allows sizing.
D	Correct	Inline elements ignore width and height.

7. Answer D Concept 5

A site ~~hosted on a \$6-a-month shared plan~~ has the page `/products/shoes.html` and the image `/images/logo.png`. Which relative path in the page's `img` tag loads the logo?

Target word: correct relative path **Decides:** Up one folder from products, then into images.

A	Wrong stem number	Climbs two levels; the page is only one level deep.
B	Confusion set	A local disk path, not a web path.
C	Dropped element	Looks for <code>/products/images/logo.png</code> .
D	Correct	<code>..</code> leaves products; then <code>images/logo.png</code> .

8. Answer D Concept 5 • transfer

The page `/blog/2026/post.html` contains an image with `src ../img/a.png`. Which file does the browser request?

Target word: resolve the path **Decides:** `..` from `/blog/2026/` is `/blog/`.

A	Adjacent category	Climbs to the root; <code>..</code> goes up only one level.
B	Absolute language	<code>..</code> exists to go up.
C	Dropped element	Ignores the <code>../</code> .
D	Correct	Up one folder to <code>/blog/</code> , then <code>img/a.png</code> .

9. Answer B Concept 6

A login form for a clinic portal has a blue submit button and fields for username and password. The site uses HTTPS. Which method should the form use, and why?

Target word: form method for credentials **Decides:** GET places data in the URL.

A	Confusion set	Speed is not the issue; GET exposes data in the URL.
B	Correct	POST sends data in the request body.
C	Absolute language	They send data in different places.
D	Adjacent category	HTTPS encrypts transit, but the URL still lands in history and logs.

10. Answer C Concept 7

~~Analytics from last quarter show most visitors to a recipe site use phones.~~ The designer wants a one-column layout only on screens 600px wide or narrower. Which media query fits?

Target word: query for small screens **Decides:** “600px or narrower.”

A	Inverted operation	Applies at 600px and above.
B	Adjacent category	Targets printing, not screen size.
C	Correct	Applies at 600px and below.
D	Confusion set	Fixes a width; it is not conditional.

11. Answer A Concept 7 · transfer

A stylesheet has @media (min-width: 768px) { .sidebar { display: block; } } and .sidebar is hidden otherwise. At which window width is the sidebar shown?

Target word: width that matches **Decides:** min-width means at or above 768px.

A	Correct	$1024 \geq 768$.
B	Absolute language	Media queries are plain CSS.
C	Inverted operation	Treats min-width like max-width.
D	Confusion set	min-width is a floor, not an exact match.

12. Answer B Concept 8

A nonprofit’s annual report page shows a ~~400-KB-PNG~~ bar chart revealing that donations rose 20% from 2025 to 2026. The chart is the only place this fact appears. Which alt text is best?

Target word: alt for an informative image **Decides:** The image carries information not found elsewhere.

A	Adjacent category	title is not a reliable text alternative.
B	Correct	Conveys the information the image holds.
C	Confusion set	Empty alt is for decorative images only.
D	Dropped element	A file name says nothing about the content.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	6	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	8	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	4	
Absolute language	Always, never, guarantee — where judgment applies.	5	
Confusion set	Every option is real; the test is exact role-matching.	11	
Camouflage	Answers a stem fact that was never the question.	0	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	CSS specificity	ID selectors beat class selectors, which beat element selectors; source order only breaks ties of equal specificity.	Thinking the last rule in the file always wins.	Several rules target the same element with different selector types.
2	The box model	With content-box (the default), rendered width = width + left/right padding + left/right border; margin is outside. With border-box, width includes padding and border.	Forgetting padding and border, or adding margin to the element's width.	width, padding, border, and margin values in one rule.
3	Semantic HTML	Use elements that describe meaning: nav for main navigation, header, main, article, footer.	Wrapping everything in div because it looks the same.	"Which element best describes..."
4	Block vs. inline elements	Inline elements (span, a) ignore width and height; change display to block or inline-block to size them.	Adding !important to force a width on an inline element.	A width that "does nothing" on span or a.
5	Relative paths	A relative path starts from the current page's folder; ../ moves up one folder.	Leaving out ../ or using a local disk path.	A page in a subfolder linking to a file in a sibling folder.
6	Form methods	GET puts form data in the URL; POST sends it in the request body, so use POST for passwords and data changes.	Thinking HTTPS makes GET fine for passwords.	A login or payment form and the method attribute.
7	Media queries	max-width applies styles at that width and below; min-width applies at that width and above.	Swapping min-width and max-width.	"Only on small screens" or "only on wide screens."
8	Accessible images	Informative images need alt text that conveys their meaning; purely decorative images get an empty alt.	Using empty alt or a file name for an image that carries information.	An image that shows data or content a user needs.

Course scope (public description): Welcome to Web Development Foundations! In this course you will learn about web design and development using HTML and CSS. This course employs the zyBooks learning platform which contains all the necessary reading materials, knowledge checks, and hands-on activities to prepare you for the course assessment. For the best understanding of the course content, complete each chapter. There are no prerequisites. Competency will be demonstrated with a performance assessment.